

GALNOEL PETER RINDENGAN

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SUMMARY PROFILE

Informatics Engineering graduate with hands-on experience designing, building, and deploying production-grade software and AI systems, recognized as a multi-year GEMASTIK Data Mining Finalist and PIMNAS 38 Bronze Medalist. Experienced across the full engineering lifecycle: backend architecture and database design (NestJS, PostgreSQL, Redis), CI/CD and observability (Docker, MLflow, Grafana, Prometheus), and applied machine learning (LLM fine-tuning, RAG pipelines, deep learning classification). Delivered an on-premise RAG assistant for PT. Bank SulutGo, rated 95/100 for discipline and 90/100 for presentation skills by the field supervisor. Strong foundation in Python, TypeScript, and end-to-end system design.

EDUCATION

Universitas Sam Ratulangi, Manado, Indonesia

Bachelor of Computer Science (S.Kom) in Informatics Engineering | CGPA : 3.93/4.00

2022 - 2026

- **Relevant Courses:** Artificial Intelligence, Data Structures & Algorithms, Database, Probability and Statistics, Software Engineering.

AWARDS

- 9th Place 100xDevs Frontier Hackathon Track August 2026
- 3rd Place Presentation Category in PKM KC at Pekan Ilmiah Mahasiswa Nasional (PIMNAS) Ke-38 November 2025
- Most Inspiring Team in GEMASTIK XVIII October 2025
- Finalist of Data Mining Division in GEMASTIK XVII October 2024
- 1st place in SATRIA DATA 2024 selection at UNSRAT level for Big Data Challenge May 2024

PROJECTS

Gamified Exam Prep Platform (YUDHA) – Backend Developer

Apr 2026 – Aug 2026

- Built a server-authoritative card battle engine using NestJS and Socket.IO, deploying Upstash Redis Lua scripts for target-aware FIFO matchmaking, private room codes, and 30-second disconnect recovery.
- Implemented transactional PostgreSQL RPCs in Supabase for idempotent match resolution, combo-scaled damage/heal mechanics, rating updates, and season progression without exposing direct table writes.
- Developed REST endpoints for question session locking and deterministic performance analytics, paired with a NestJS background scheduler integrating Firebase (FCM) for streak-rescue push notifications.

Reasoning-Driven Legal AI Assistant

July 2026

- Fine-tuned Qwen 2.5 7B using QLoRA and implemented Group Relative Policy Optimization (GRPO) with custom multi-objective reward functions to induce <think> chain-of-thought reasoning for Indonesian labor law compliance.
- Engineered a multi-stage Advanced RAG pipeline integrating HyDE (Hypothetical Document Embeddings), Parent-Child hierarchical chunking, and a Hybrid Ensemble Retriever (BM25 + ChromaDB RRF) across 1,900+ legal document pages.
- Deployed a Cross-Encoder Reranker (FlashRank) with confidence threshold gating and DuckDuckGo fallback, serving merged 16-bit model artifacts published on Hugging Face Hub via a Gradio interface.

End-to-End MLOps & Real-Time Model Observability Pipeline

June 2026

- Architected an automated MLOps CI/CD pipeline using MLflow, Docker, and GitHub Actions to execute automated unit testing, model artifact versioning, and zero-touch image publishing to Docker Hub.
- Built high-performance model serving telemetry using a custom Prometheus exporter, tracking real-time inference throughput, request error rates, and p95 latency.
- Provisioned automated Grafana observability dashboards with multi-tier alerting rules to proactively trigger incident notifications on latency spikes and service degradation.

Solana-Based Web3 Game (CORA) – Backend Developer

Apr 2026 – May 2026

- Engineered a high-throughput backend system using Bun, Hono, and Supabase, orchestrating real-time WebSocket state management and matchmaking for an on-chain aptitude wagering arena.
- Programmed an automated server-side cryptographic settlement oracle utilizing @solana/web3.js, securely executing deterministic payload signatures to verify match outcomes and trigger automated token payouts.
- Built fully compliant Solana Actions API endpoints, enabling automated wSOL wrapping for native SOL wagers, and implementing a sequential deposit state machine to prevent race conditions.
- Integrated the backend with MagicBlock Ephemeral, enabling a verifiable, on-chain mirror of real-time gameplay loops to provide decentralized fairness proofs.

Multimodal Neurotherapeutic Toolkit (NEUROKIT) – Project Leader

Jul 2025 – Nov 2025

- Engineered an ESP32-based therapeutic device using an HC-SR04 ultrasonic sensor and PWM-regulated MOSFET circuitry to dynamically adjust LED brightness, maintaining an optimal 500–600 lux (536 lux avg) across user distances of 30–70 cm.
- Developed a Flutter mobile application connected via Bluetooth to control dual-mode audio therapy (Learn/Chill), gamify study sessions, and generate personalized audio affirmations using OpenRouter LLM and Prosa TTS APIs.
- Conducted functional trials using the Wechsler Memory Scale (WMS), achieving an 11.35% improvement in student attention span and memory retention, which secured 3rd Place (Bronze Medal) in PKM-KC at PIMNAS 38.

Javanese Aksara Image Classification (GEMASTIK XVIII Finalist)

October 2025

- Engineered a fine-grained classification pipeline using CoaT (Co-Scale Conv-Attentional Transformers) to capture multi-scale stroke representations across 20 complex script classes.
- Formulated a two-stage training strategy combining frozen patch-embedding feature extraction and progressive fine-tuning with AdamW and Kornia Focal Loss ($\gamma=2.0$, $\alpha=0.25$) to mitigate severe class imbalance.
- Implemented Explainable AI (XAI) using Captum Integrated Gradients to generate pixel-level attribution heatmaps, verifying stroke-level attention and eliminating shortcut learning.
- Achieved a 96.7% test Macro F1-score across validation and final competition evaluation sets.

Multimodal Fish Freshness Classification (GEMASTIK XVIII)

Jul 2025 – Aug 2025

- Multimodal Data Synchronization: Curated and aligned visual imagery and e-nose gas sensor streams from the DaFiF dataset using nearest-neighbor timestamp matching, Z-score anomaly filtering, and edge-detection augmentations.
- Contrastive & Fusion Architecture: Engineered a two-stage deep learning pipeline combining a ViT (DINOv2 backbone) and MLP sensor encoder aligned via CLIP-style contrastive loss, integrated through bidirectional parallel cross-attention transformer layers.
- Optimized training via Focal Tversky Loss to achieve 85.4% accuracy and an 84.3% macro F1-score (peaking at 92.7% on Marginal class).

AI Math Tutor Web App (Socratix) – AI Engineer

Apr 2025 – May 2025

- Multi-Agent Architecture: Designed and implemented a two-tier micro-agent pipeline (Router & Planner) using Groq and NestJS, optimizing response workflows for an automated Socratic math tutoring platform.
- Agent #0 (Router): Engineered a high-efficiency intent classifier combining regex fast-paths with LLM fallbacks via Vercel AI SDK, routing student inputs within ~5ms and achieving 100% classification accuracy on a 20-sample validation matrix.
- Agent #1 (Planner): Developed a structured context extraction engine utilizing Zod schema validation and JSON-only prompting, successfully capturing complex equations, problem subtypes, and multi-modal parameters while enforcing prompt-injection mitigations.
- Reliability & Resilience: Architected defensive error-recovery layers and graceful degradation protocols, maintaining system stability and eliminating down-stream validation failures across edge-case user queries.

EXPERIENCE

AI Engineer Intern

Jun 2025 – Aug 2025

PT. Bank SulutGo, Manado, Indonesia

- RAG Chatbot Development: Architected and deployed an on-premise virtual assistant using Ollama, n8n, and Supabase (pgvector) to eliminate internal "information silos".
- Data Sovereignty: Ensured 100% data security by configuring the entire AI stack (LLM & Vector DB) to run locally on bank servers, preventing data leakage to third-party cloud services.
- System Design: Authored comprehensive technical documentation, including Use Case, Sequence, and Entity Relationship Diagrams (ERD) to map the interaction between the Next.js frontend and the AI backend.
- High Evaluation: Achieved a near-perfect 95/100 score in "Discipline and Attitude" and 90/100 in "Presentation Skills" from the field supervisor.

Development Lead Chatbot Project for TECHOFEST 2024

Jul 2024 – Nov 2024

UNSRAT IT Community, Manado, Indonesia

- Directed a 12-person multidisciplinary engineering team to build and deploy an AI chatbot web and mobile application leveraging Gemini AI for the faculty's student organization.
- Deployed model to server with flask using GRPC to help serialize (organize) the data for fast transmission between services.
- Coordinated the software team in the end-to-end development of accompanying mobile and web applications.
- Supervised the data engineering team, overseeing data collection workflows and analysis pipelines.

Committee of Machine Learning Division

Jan 2024 – Jul 2024

UNSRAT IT Community, Manado, Indonesia

- Designed a comprehensive 6-month curriculum covering fundamental to advanced machine learning concepts.
- Instructed and facilitated workshops for a cohort of 20+ members passionate about AI and Machine Learning.
- Mentored a 5-person junior team through the full lifecycle of a chatbot development project.

SKILLS

Programming Language: Python, TypeScript, JavaScript

AI/ML: PyTorch, TensorFlow, Hugging Face, Ollama

Backend & Infrastructures: NestJS, Flask, Docker, MLflow, Redis, Supabase (pgvector), n8n, Git, Grafana, Prometheus

Databases: PostgreSQL, MySQL, MongoDB, MariaDB.

Language : Indonesian (Native) and English (Intermediate)